

1. Objectives:

1. To understand loops and conditional statements in Java
2. To understand the switch statement and logical conditions in java.

2. Introduction:

Java is a high-level, object-oriented programming language that provides different control structures to control the flow of a program. In Java, loop statements such as for, while, and do-while are used to execute a block of code repeatedly until a specific condition becomes false. These loops are commonly used in tasks such as pattern printing, calculations, and iterative operations. Decision-making statements such as if-else and switch are used to check conditions and perform different actions based on the result of those conditions. The if-else statement evaluates logical expressions, while the switch statement selects one block of code from multiple options based on a variable's value. By using loops and conditional statements together, programmers can create programs that perform logical operations, control program flow, and solve practical problems efficiently.

3. Required Hardware and Software:

- **Hardware:** AMD RYZEN 7, RAM 8 GB
- **Software:** Visual Studio Code 1.109

4. Experiment Implementation:

Experiment 1: Pattern Printing

```
package ThirdLab;

public class ClassOne {

    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= 5; j++) {
                if (i == 1 || j == 1 || i == 5 || j == 5) {
                    System.out.print("* ");
                } else {
                    System.out.print("  ");
                }
            }
            System.out.println();
        }
    }
}
```

3.1: Experiment 1 output

```
PS D:\BAUST CSE\Java Programing\3rd_Lab> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe'
'-cp' 'C:\Users\Rezwan\AppData\Roaming\Code\User\workspaceStorage\949bbfa8f628b19c3a3f99e
ClassOne'
* * * * *
*       *
*       *
*       *
*       *
* * * * *
PS D:\BAUST CSE\Java Programing\3rd_Lab> █
```

Experiment 2: Switch Case

```
package ThirdLab;

import java.util.Scanner;

public class ClassTwo {
```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number: ");
    int n = sc.nextInt();

    switch (n) {

        case 1:
            System.out.println("One");
            break;
        case 2:
            System.out.println("Two");
            break;
        case 3:
            System.out.println("Three");
            break;
        default:
            System.out.println("Not a number");
            break;
    }
    sc.close();
}
}

```

3.2: Experiment 2 output

```

PS D:\BAUST CSE\Java Programing\3rd_Lab> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe'
'-cp' 'C:\Users\Rezwan\AppData\Roaming\Code\User\workspaceStorage\949bbfa8f628b19c3a3f99e
ClassTwo'
Enter number: 7
Not a number
PS D:\BAUST CSE\Java Programing\3rd_Lab>

```

Experiment 3: Check Prime Number

```

package ThirdLab;

import java.util.Scanner;

public class ClassThree {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int count = 0;
        for (int i = 2; i < num; i++) {
            if (num % i == 0) {
                count++;
            }
        }

        if (count == 0 && num > 1) {
            System.out.println("Prime Number");
        } else {
            System.out.println("Not Prime Number");
        }

        sc.close();
    }
}

```

3.3: Experiment 3 output

```
PS D:\BAUST CSE\Java Programing\3rd_Lab> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe'
'-cp' 'C:\Users\Rezwan\AppData\Roaming\Code\User\workspaceStorage\949bbfa8f628b19c3a3f99e
ClassThree'
Enter a number: 12
Not Prime Number
PS D:\BAUST CSE\Java Programing\3rd_Lab> 
```

5. Discussion & Conclusion:

In this lab, we have learned loops. We have used for loop was learned to perform pattern printing. Decision-making techniques were also practiced using conditional statements and the switch statement to control program flow based on different conditions. Additionally, the Scanner class was used to take input from the user, and checking whether a number is prime, theoretical concepts were successfully applied in practice. These exercises helped strengthen understanding of loops, conditional logic, and problem-solving techniques in Java programming, which are essential for developing more advanced programs in the future.

7. Reference:

1. <https://www.geeksforgeeks.org/java/java-prime-number-program/> accessed at 7.35 PM on 7th March 2026.